

Chapter 3: Linear regression

Introduction to Quantitative Methods

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2024

R - Quick presentation

- **R is a coding language specialised in the manipulation of dataframes**
 - User friendly, and very powerful
 - Another famous data-handling language is Python: more polyvalent than R, but not as intuitive for data-handling
 - If you know how to use R, you can easily learn Python on your own
 - You might also hear about Stata, SAS or SPSS, but these languages are getting "old-school"
 - C or C++ are for the big guys, it is the language that codes R !
- Why use a coding language? (and not a ready-built, button-clicking software)
 - Flexibility
 - Replicability
 - Efficiency
- Why do most people not know how to code?
 - 50% they lacked a proper introduction
 - 50% they think code is scary

- R is a coding **language**
- It needs an **environment** (an interpreter). The most popular one is **RStudio**
- R is a language that has been expanded by people. They created **packages** or **libraries**
 - For instance, base R does not allow to make good-looking graphs. So we will need to install the ggplot2 library
- To use such packages, you need to install an application called **Rtools**

To be fully operational in R, you therefore need to download **R**, **RStudio**, and **Rtools**

Equations and functions

Equation: A relationship between different variables

Function: The formal relationship that links variables together

Example:

- Let us express a mass in kilos with the variable y .
- Let us express a mass in grams with the variable x .
- We have the following equation: $y = 1000 \times x$
- We can build the following function:
- $f : x \rightarrow f(x) = 1000 \times x$
- f is defined as the relationship that transforms x into $f(x)$, where $f(x) = 1000 \times x$

Some types of functions are more commonly used than others:

- **Affine (linear) functions:** $f(x) = a \times x + b$
- **Square function:** $f(x) = x^2$
- **Polynomial function:** $f(x) = a \times x^2 + b \times x + c$
- **Logarithmic function:** $f(x) = \log(x)$

And many others: inverse function, square root function, trigonometric functions, exponential function...

If you need help to visualise these relationships, I recommend plotting them: [▶ Link](#)

Functions can become more complex, the more complex you want your relationship to be.

Example:

- You earn x from your part-time job, y from your own company and z from babysitting. You spend w on your rent and v on your food.
- Your disposable income can be represented by the function g where
- $g : x, y, z, w, v \rightarrow g(x, y, z, w, v) = x + y + z - w - v$
- Now the government taxes your company at a rate of $a\%$ and subsidises your rent by $b\%$:
- $g : x, y, z, w, v, a, b \rightarrow g(x, y, z, w, v, a, b) = x + (1 - a) \times y + z - w - (1 - b) \times v$

Linear regression - Definition and intuition

- A linear regression is a statistical method that determines **the best linear relationship** between variables
- **The best** means that the relationship does not have to be exact!
- It allows us to compute:
 - Intensity of relationships, effect sizes
 - Probabilities that observed relationships are not random
 - Sometimes, causality

Formal representation:

- We study the relationship between a **dependent variable** (the outcome) Y and the **explanatory variables** $X_1, X_2, X_3 \dots$
- $Y = a + b_1 \times X_1 + b_2 \times X_2 + b_3 \times X_3 + e$
- a is the **intercept**
- b_1, b_2, b_3 are the **regression coefficients** of the variables X_1, X_2, X_3
- e is called the **error term**
- e is a variable (like X and Y) whereas a is a constant (like $b_1, b_2 \dots$)

$$Y = a + b_1 \times X_1 + b_2 \times X_2 + b_3 \times X_3 + e$$

How to read this thing:

- We say that **we regress Y on X_1 , X_2 and X_3**
- The intercept is very rarely commented
- The regression coefficients give you the sign and the size of the relationships
- The error term is (or rather, should be) random.
- The error term is always **on average** equal to 0.
- The more the values of the error term are close to 0 (the smaller the variance), the better the fit
- The fit of a regression is *how well* the regression explains the dependent variable (like a shoe fits your foot)

What is the difference between this (*):

$$Y = a + b_1 \times X_1 + b_2 \times X_2 + b_3 \times X_3 + e$$

And this? (+):

- $Y = a_1 + b_1 \times X_1 + e_1$
- $Y = a_2 + b_2 \times X_2 + e_2$
- $Y = a_3 + b_3 \times X_3 + e_3$

Answer:

- b_1 , b_2 and b_3 are (almost never) equal in the contexts (*) and (+)!!
- We say that X_2 and X_3 **control** X_1 (or that X_1 and X_3 control X_2 etc..)

Here we have a very simple linear regression model (these numbers are 100% fake):

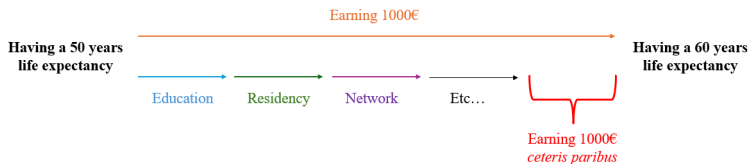
- Let's say earning 100€ more increases by 1 year your life expectancy
- $Y = 50 + 0.01 \times X + e$



- We may notice in our model that e is on average very small. This means that income explains a lot how long we are expected to live.
- However, this model has one fundamental issues. Is it income, or *something else associated to income* that explains how long we live?
- Income may have an impact on how much we are able to spend on healthcare
- But our income is also associated to our education, where we live, our social network, the type of job we have...
- **We therefore want to control for these mechanisms to measure the effect of income**
- **For a same** education, same place of residency, same social network, same work penibility... what is the effect of earning 100 more euros on your health expectancy?
- **All other things equal** (*ceteris paribus*) what is the effect of earning 100 more euros on your health expectancy?

We introduce other variables inside the regression (still fake numbers):

- Let's say this time X_1 measures income and $X_2, X_3...$ are our control variables
- $Y = 50 + 0.005 \times X_1 + b_2 \times X_2 + b_3 \times X_3 + ...e$
- In this model, we do not care about b_2 and b_3 (in order to analyse them we would have to control them too)
- The effect of income has become smaller, because some of the effect of income was **captured** by the controls



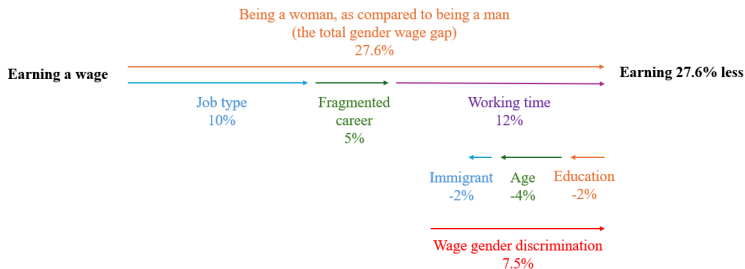
Some famous application of such decomposition are studies measuring gender (or racial) wage discriminations (I know this is a very sensitive topic so feel free to ask questions on or debate the contents of these slides): [▶ Link](#)

- We observe in most OECD countries a negative relationship between being a woman (as compared to being a man) and wage, 27.6% in Switzerland
- But women are working in different **types of jobs** than men. If we control for this does this relationship stand? $\approx 17\%$
- But women are **working less** than men. $\approx 5\%$
- But women have more **fragmented careers**. $\approx 0\%$ (no more wage gap)

However, controlling goes both ways!

- Women are on average more **educated** than men. $\approx 2\%$
- Women are on average **older** (healthier and more experienced) than men. $\approx 6\%$
- Women are on average less **immigrants** than men (we are not looking at discrimination intersectionalities). $\approx 7.5\%$
- Hence, in Switzerland, for a similar job and competences, women are still payed $\approx 7.5\%$ less than men

- We say that $\approx 7.5\%$ is the unexplained (by the controls) effect of being a woman on her wage
- It does not mean that we cannot explain it (we deduce that it is discrimination)
- It does not mean either that what we control is not some other form of discrimination



Confidence intervals, significance

An interval: A pair of values (an upper and lower bound) that bound a variable

- Let's say we have a distribution X of values $x_1, x_2, \dots, x_N$
- The maximum and the minimum are the empirical bounds of a variable: $x_i \in [\min(X), \max(X)]$
- 0 and infinity are the theoretical bounds of GDP, of height or of age: $GDP \in [0, \infty[$

The confidence interval: An interval associated with a probability (a confidence)

- Usually, we use the confidence of 90%, 95% or 99%
- 95% of people are between 2.0 and 1.4m tall:

$$P(\text{height} \in [1.4, 2]) = 0.95$$
- You have 80% chance to win nothing when you play loto:

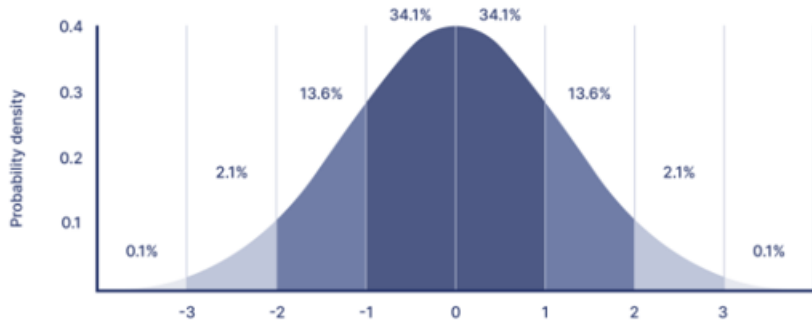
$$P(\text{gain} \in [0, 0]) = P(\text{gain} = 0) = 0.80$$
- An archer hits bullseye 50% of the time: $P(\text{arrow} = \text{bullseye}) = 0.50$

There can be several confidence intervals with a similar confidence!

- You have 50% chances to win at head-or-tail, and 50% chance to lose
- $P(x_i \in [\min(X), \text{med}(X)]) = P(x_i \in [\text{med}(X), \max(X)]) = 0.5$
- But at given confidence interval, **only one** is centered at the median!

$$: P(x_i \in [\text{quartile}_1(X), \text{quartile}_3(X)]) = 0.5$$
- And **only one** is centered around the mean, harder to compute though

Standard normal distribution

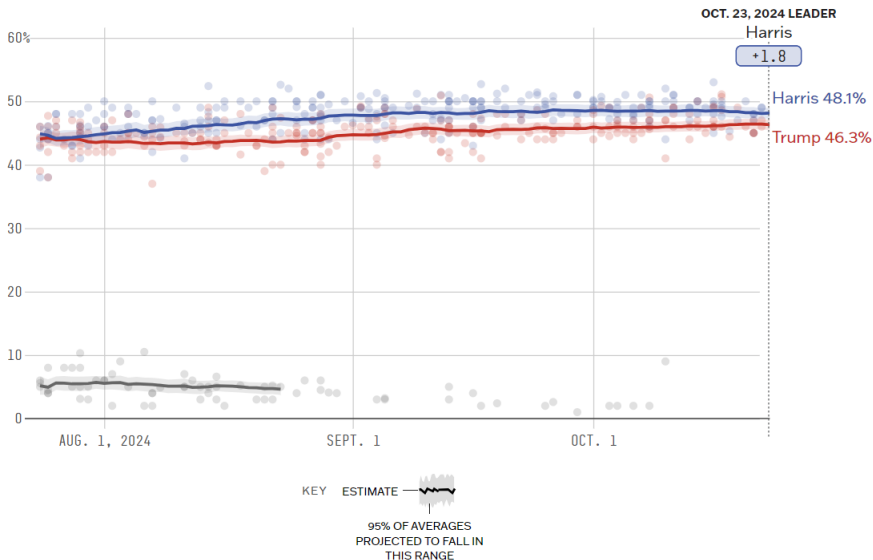


DATES	SAMPLE ⓘ	POLLSTER	SPONSOR	RESULT				NET RESULT
President: general election, 2024 ↗ AVG.								
Oct. 20-22	1,294 LV	TIPP Insights		Harris	49%	47%	Trump	Harris +2
Oct. 19-22	1,293 LV	YouGov	The Economist	Harris	49%	46%	Trump	More ▼ Harris +3
Oct. 19-22	1,422 RV	<u>YouGov</u>	The Economist	Harris	48%	46%	Trump	More ▼ Harris +2
Oct. 21	1,273 LV	Redfield & Wilton Strategies		Harris	45%	47%	Trump	Trump +2
Oct. 18-21	1,189 LV	YouGov	The Times of London, SAY24	Harris	48%	45%	Trump	More ▼ Harris +3
Oct. 18-21	1,266 RV	YouGov	The Times of London, SAY24	Harris	47%	45%	Trump	More ▼ Harris +2
Oct. 17-21	802 RV	Monmouth University Polling Institute		Harris	47%	44%	Trump	Harris +3
Oct. 16-21	3,307 LV	Ipsos	Reuters	Harris	47%	45%	Trump	More ▼ Harris +2
Oct. 16-21	3,481 RV	Ipsos	Reuters	Harris	45%	43%	Trump	More ▼ Harris +2
Oct. 16-21	4,129 A	Ipsos	Reuters	Harris	41%	39%	Trump	More ▼ Harris +2
Oct. 16-21	4,129 A	Ipsos	Reuters	Harris	41%	39%	Trump	Harris +2
Oct. 16-21	3,307 LV	Ipsos	Reuters	Harris	48%	45%	Trump	Harris +3
Oct. 16-21	3,481 RV	Ipsos	Reuters	Harris	46%	43%	Trump	Harris +3
Oct. 18-20	8,570 LV	Morning Consult		Harris	50%	46%	Trump	Harris +4



Who's ahead in the national polls?

Updating average for each candidate in 2024 presidential polls, accounting for each poll's recency, sample size, methodology and house effects.



Result significant at the $x\%$ level: When we are $x\%$ sure that the result is true

Value significantly greater/smaller from y at the $x\%$ level: We can build an $x\%$ confidence interval in which y is not part of

- Someone's height is significantly greater than 1.4m at 95% : 95% of the time, people are taller than 1.4m

An effect is significant: *Abus de langage*. When a correlation is significantly greater/smaller than 0 at least at the 90% level

- Per our available data, gender discrimination is significant
- In 23/10/2024, Harris significantly wins the US popular votes

When do we use confidence intervals?

- When we do not measure everything (when we have a sample)
 - A sample is a random pick of observations
 - We could have picked other people and maybe got another result
 - So we need to compute the probability that what we observe is true
- When we observe everything but not every possible events
 - It is not because we observe all our population that we are certain that their behaviour would be the same if we world would repeat exactly itself
 - We observe the height of people in a village. We are unsure if what we observe is purely random (because of the genetic shuffle) or because of some characteristics specific to the village
 - Far-right movements are gaining momentum, is it a succession of nation-specific occurrences, or is it a world phenomenon?

When do we not use confidence intervals?

- We do not measure confidence intervals when we measure everything and all possible events. For instance:
 - You have a census of the entire population. You do not have to say the median is more or less... **because** you know it for certain that you measured everything
 - You observe all countries of the world and you observe a correlation between GDP and life expectancy. That correlation is the true one
 - NB: In this scenario, you may want to check if this relationship is robust to controls such as health expenditures, education, etc.
 - NB: That's because you do not observe exactly all possible configurations of countries (high GDP, low expenditure, high education / medium GDP, high expenditure, low education...)

We usually build confidence intervals on...

Any descriptive statistic:

- The mean, the median, the variance, the maximum etc.

Multivariate statistics:

- The covariance, the correlation coefficient, the **linear regression coefficient**

We have seen previously that we can distinguish correlation from causality. **But causality can only be concluded if the correlation we observe is significant (not random).**

Implementing a linear regression

$$RN\ vote\ share = a + b \times Cadres + e$$

```
Call:
lm(formula = RN_share ~ cadres, data = reg_data)

Residuals:
    Min       1Q   Median       3Q      Max
-11.5169  -3.4898  -0.0232   3.3076  11.0703

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  26.60402    1.32948   20.011  < 2e-16 ***
cadres       -0.50311    0.09031   -5.571  2.4e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.435 on 94 degrees of freedom
(11 observations deleted due to missingness)
Multiple R-squared:  0.2482,    Adjusted R-squared:  0.2402
F-statistic: 31.03 on 1 and 94 DF,  p-value: 2.402e-07
```

$$RN\ vote\ share = 26.6 - 0.50 \times Cadres + [-12, 11]$$

$$P(b < 0) = 1 - 2.4 \times 10^{-7} = 0.00000024$$

Regression
formula

Intercept

Linear
coefficient

```
Call:
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```

Statistics on e

Probability
that $b \neq 0$

R², how much
we explain Y

$$RN\ vote\ share = a + b_1 \times Cadres + b_2 \times Median\ income + e$$

```
Call:
lm(formula = RN_share ~ cadres + disp_medyy, data = reg_data)

Residuals:
    Min       1Q   Median       3Q      Max
-15.5286  -3.3677  -0.2445   3.8262  11.8371

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  67.6311152    7.9224400   8.537 5.08e-13 ***
cadres        0.1848198    0.1537610   1.202  0.233
disp_medyy   -0.0024296    0.0004617  -5.262 1.07e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.847 on 84 degrees of freedom
(20 observations deleted due to missingness)
Multiple R-squared:  0.4346,    Adjusted R-squared:  0.4211
F-statistic: 32.28 on 2 and 84 DF,  p-value: 3.982e-11
```

$$P(b_1 > 0) = 1 - 23\%$$

$$P(b_2 < 0) = 1 - 1.07 \times 10^{-6} = 0.00000107$$

```
Call:
lm(formula = RN_share ~ cadres + disp_medyy, data = reg_data)

Residuals:
    Min       1Q   Median       3Q      Max
-15.5286  -3.3677  -0.2445   3.8262  11.8371

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P-value: Statistic giving the $1 - \text{probability}$ that the linear regression coefficient is smaller or greater than 0.

- You read it as the probability that the correlation does not exist
- So $1 - p\text{-value}$ is the probability that the correlation does exist
- There are usually stars next to it. The more stars, the more the coefficient is significant!
- Finding these stars is intrinsically part of the work-reward feedback loop that drives statisticians and economists in their daily lives and careers

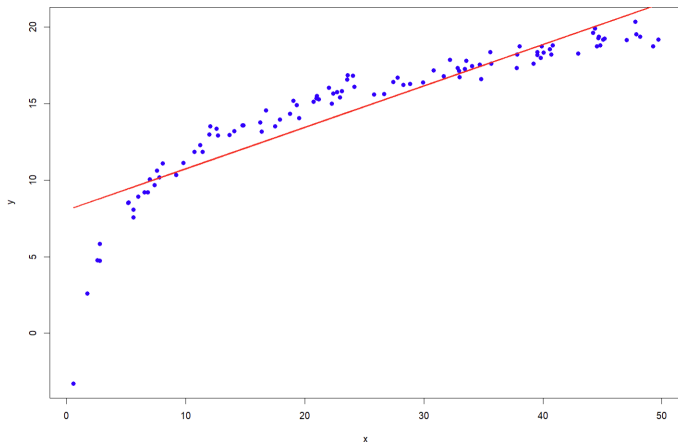
R^2 : (Intuitive definition) Share of the variance explained by the regression.

- If $R^2 = 1$, the model explains everything (everyone is on the regression line) and $e = 0$ all the time!
 - This is near-impossible to have in real life. If your R^2 is greater than 0.8, it is almost certain you have an error somewhere.
- If $R^2 = 0$, the model explains nothing and usually (not necessarily!) the p-values are > 0.5 .
- If $R^2 = x$, the model explains $100 \times x\%$ of the variations / fluctuations of Y

Error term, residual, noise: What is not explained by the model. It is very important for multiple reasons,

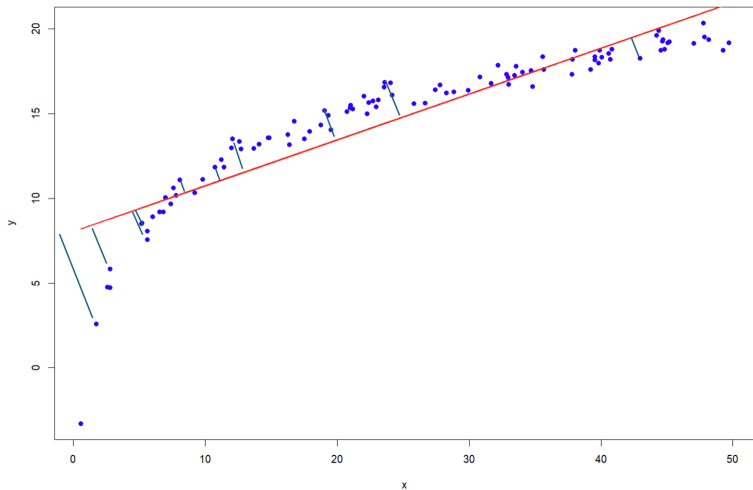
- It helps build R^2 . The closer values of e are to 0, the closer R^2 is to 1
- **It gives us information on if the relationship is non-linear**
- If the relationship is non linear, the error will not be normally distributed
- The error term is always on average equal to 0! In other words, on average, the relationship is true.
- For instance, if $E(e) = 0 \neq \text{Median}(e)$, it might suggest that the relationship we observe is not linear

$$Y = 0 + 5 \times \log(x) + e$$

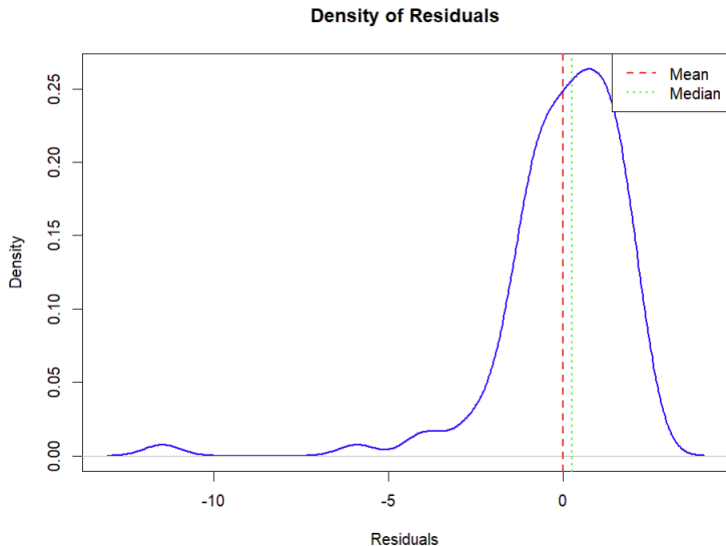


$$Y = 0 + 5 \times \log(x) + e$$

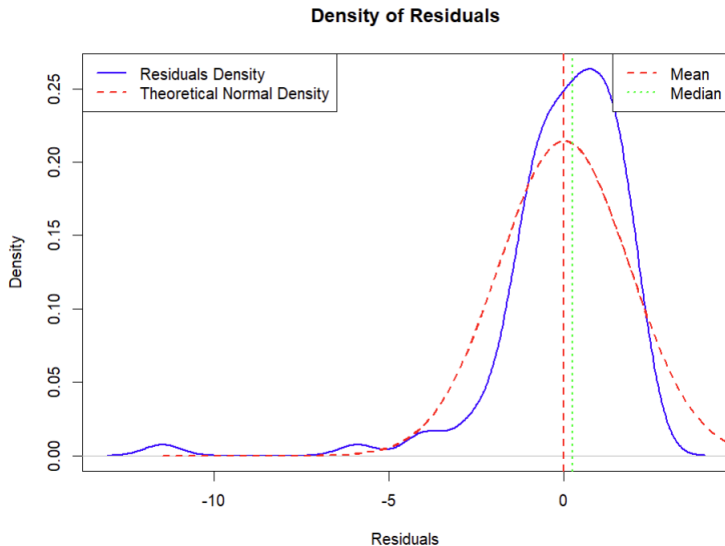
The errors e can be graphically represented the following way:



And we can represent their distribution:



The distribution is not normal, as opposed to if we had a linear relationship! :



How to understand the p-value

This material gets technical. Its sole aim is to help you understand what we are doing. For the final exam, you just need to remember the name and the associated concepts of the 2 following theorems.

The law of large numbers: Very very (very) important theorem of probability science. The more we observe identical independent observations of a variable X , the more precisely we can identify $E(X)$.

- [▶ Link](#)
- Independent means that each observation is not related to another observation whatsoever
- "More precisely" means that the confidence interval of $E(X)$ with a probability P is getting smaller
- With an infinite number of observations X_i we would be 100% sure that $\widehat{E(X)} = E(X)$
 - The $\hat{\cdot}$ symbol signifies the empirical value of a variable (as opposed to its theoretical one)
- The precision increases from the number of observation but has (very fast) decreasing returns:

$$P\left(\left|\widehat{E(X)} - E(X)\right| \geq e\right) \leq \frac{\sigma^2}{n \times e^2}$$

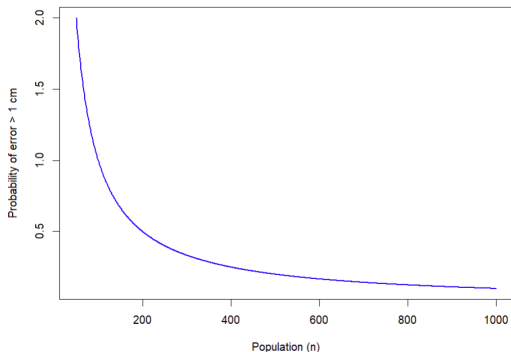
Let's study heights.

The height has a standard deviation of around $\sigma = 10cm$.

The probability to measure wrongly the mean by a margin of $e = 1cm$ is therefore:

$$P(\left|\widehat{E(X)} - E(X)\right| \geq 1) \leq \frac{10^2}{n \times 1^2} = \frac{100}{n}$$

Relationship between Population and Probability of Mean Error



So, if we want $P \leq 0.1$, we have:

$$P \leq 0.1 = \frac{100}{n} \Rightarrow n = \frac{100}{0.1} = 1000$$

And if we want $P \leq 0.01$, we have:

$$P \leq 0.01 = \frac{100}{n} \Rightarrow n = \frac{100}{0.01} = 10000$$

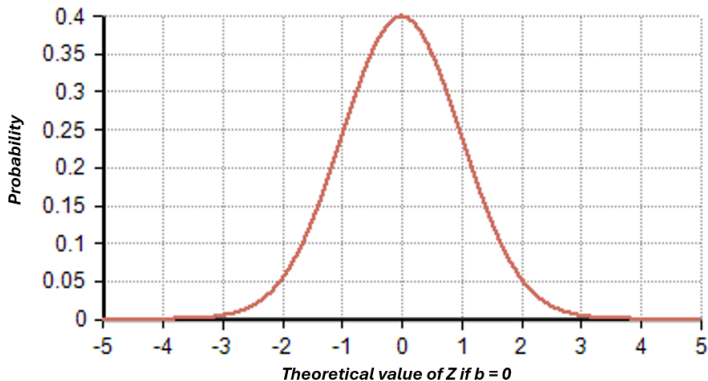
Central limit theorem: Second most important theorem of probability science. Based on the identical independent observations of a variable X , the measurement error of the mean of X has a normal distribution.

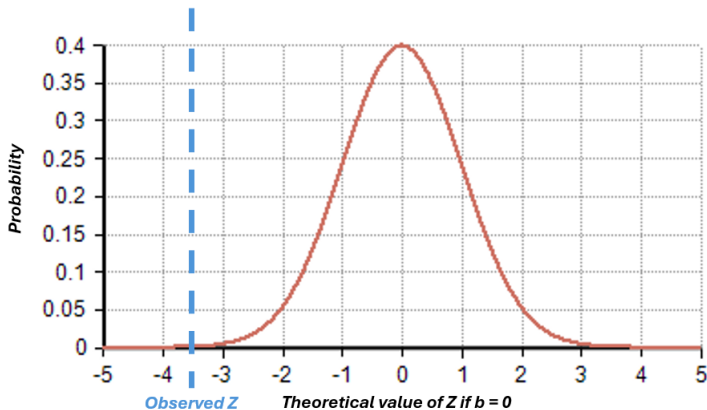
- ▶ Link
- $Z = \frac{\sqrt{n}}{\sigma^2} \times [\widehat{E(X)} - E(X)]$ for an n big enough (~ 100) has a normal distribution
- No matter what is X , you will always find that, for n big enough, Z has a normal distribution !!!
- This is why we find so often the normal distribution in nature, it is a mathematical rule just like $1 + 1 = 2$
- The law of large numbers was about the size of confidence intervals. The central limit theorem is about testing if one statistic is different from another.**

Central limit theorem: Second most important theorem of probability science. Based on the identical independent observations of a variable X , the measurement error of the mean of X has a normal distribution.

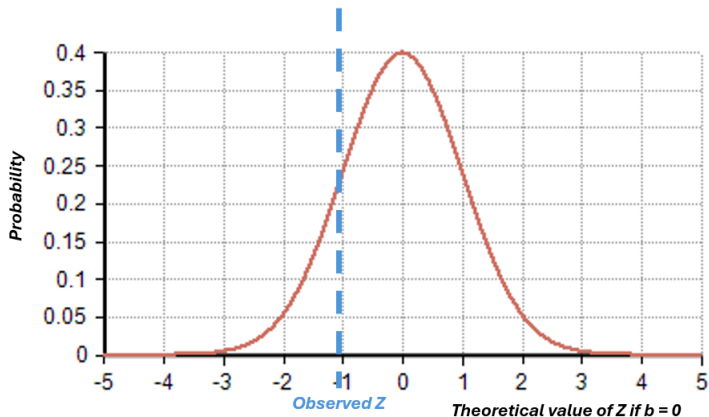
- Also, this means that we can compute the probability something (a linear regression coefficient for instance) is different from 0!
- We replace $\widehat{E(X)}$ by $\hat{b}$ and $E(X)$ by 0 and we look if Z is coherent with a normal distribution:
- $Z = \frac{\sqrt{n}}{\sigma^2} \times [\hat{b} - 0]$
- This is a wrong interpretation but it may help you understand: We look at the probability that Z is different from 0.

So, based on the theorem, we know that the distribution of Z looks like this if $b = 0$:

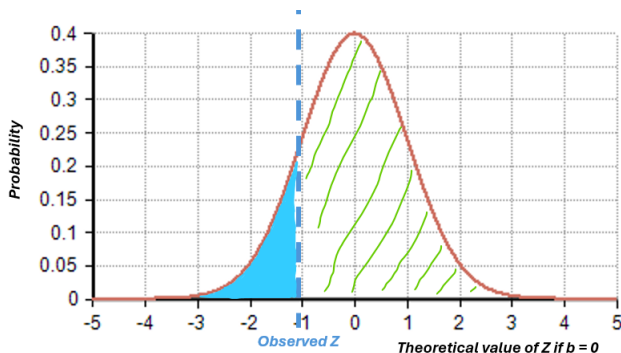




In this case scenario, the observed value of Z is very unlikely. This means that $E(X) = b < 0$ with a very high probability!



In this case scenario, the observed value of Z is somewhat possible. This means we **may** have $E(X) = b = 0$ and we cannot really conclude

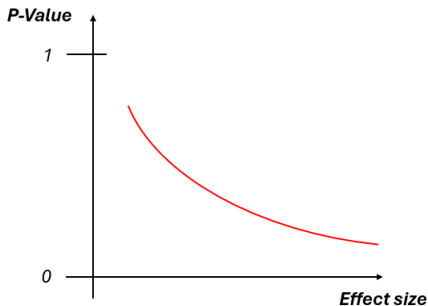
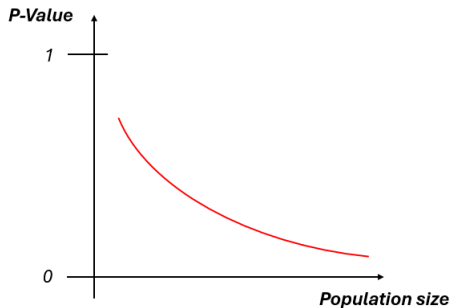


The p-value corresponds to the probability that $b < 0$, which is given by the % of the bell curve which is in under blue area.

NB: The same can be done with $b > 0$, which would be on the right handside.

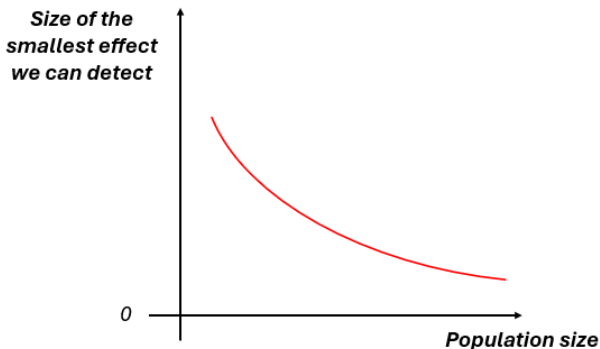
$$Z = \frac{\sqrt{n}}{\sigma^2} \times [\hat{b} - 0]$$

- We can therefore deduce some things on what makes us have a low p-value!
- The bigger the effect, and/or the bigger the number of observations, the smaller will be the p-value



Said otherwise, the more observations we have, the smaller the size of the effect we can detect!

- And the smaller the effect, the more observations we need
- But still with decreasing returns...
- even more decreasing that in the law of large numbers because of $\sqrt{n} < n$!!



Summary of what we learned so far.

We can deduce causality if:

- We observe a **significant** regression coefficient
 - its **p-value** is smaller than $0.1 = 10\%$
- We have enough **controls** to surmise *ceteris paribus*
- The **error** term is normally distributed

But that's not enough...

Issues often encountered with linear regressions and how to (maybe) overcome them

You do not identify an effect (the p-value is too big) because (not mutually exclusive):

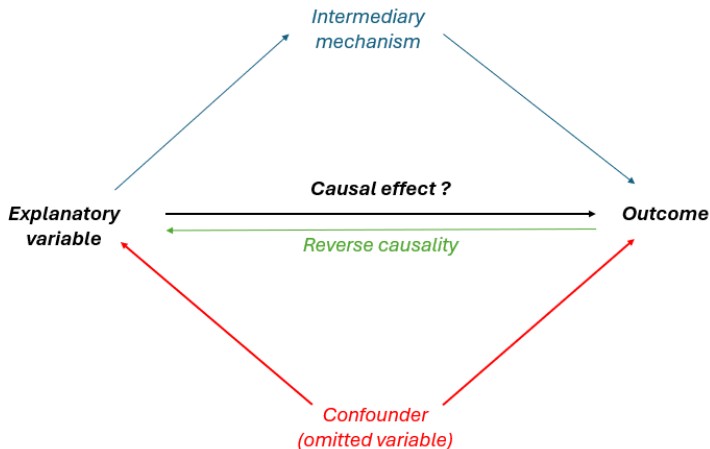
- The effect does not exist and is equal to 0
- The effect is too small to detect
- Your number of observations is too small to detect the effect
- You have too many controls
- The solution is therefore to increase the number of observations
- Or remove some controls (at the risk of losing the *ceteris paribus*)
- Or say that it is actually very interesting that the effect is not significant and may be close to 0 because bla bla bla

$$Y = a + b \times X + e$$

becomes...

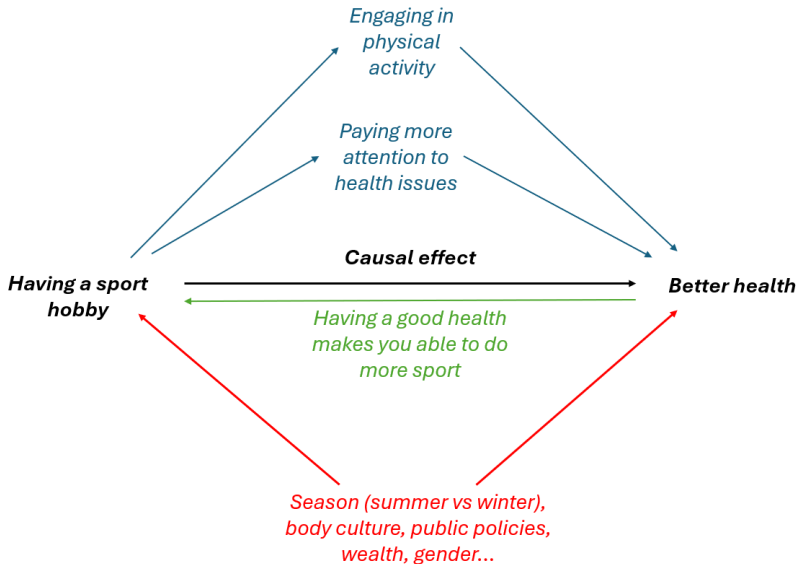
$$Y = E(Y) + 0 \times X + [Y - E(Y)]$$

Endogeneity: There is endogeneity when your model has plot holes because you could not precisely isolate the causal effect.



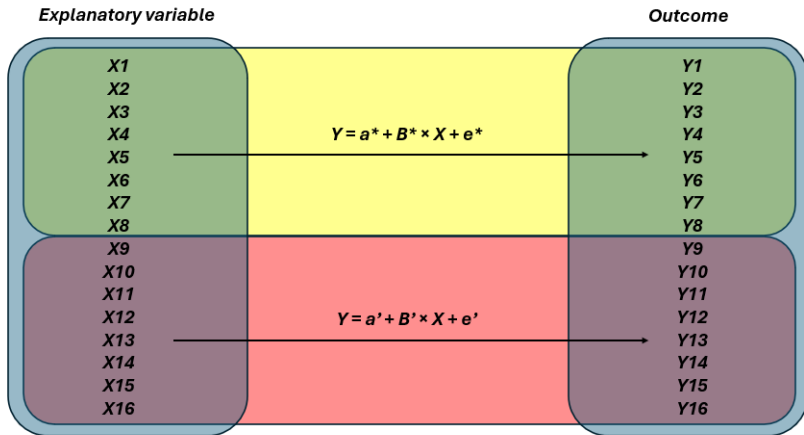
Types of endogeneity issues

- **Reverse causality:** When the causality goes both ways (for instance sports increases health, and good health incites people to have a sport hobby)
- **Confounder (unobserved variable):** When you are missing control variables, because you lack the data (or the insight)
- **Measurement error:** This is similar to an unobserved variable. This is when what you measure has been tampered with non-randomly.
 - Men tend to over-report their own health compared to women
- The only solution is to find another empirical strategy. Or being honest and admitting that you cannot establish causality
- **Intermediary mechanism:** This is not necessarily harmful, but it may change the interpretation of your results. It can actually be useful if you want to remove confounders!



Non-identical observations: you observe non comparable populations, on which the effect you want to identify is not the same (we say that the effect is heterogenous).

- The effect of access to the pill on fertility, using a population of both men and women.
- Some urban development policies will have a positive impact on the life of some neighborhoods, and a gentrification effect on others.

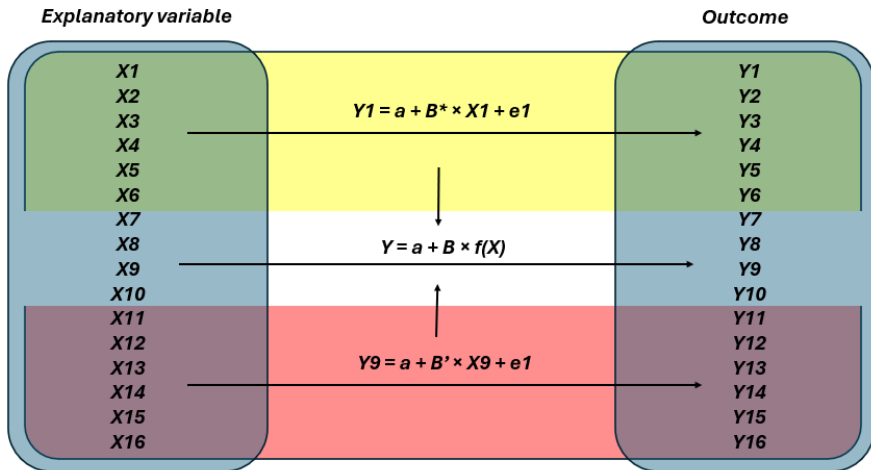


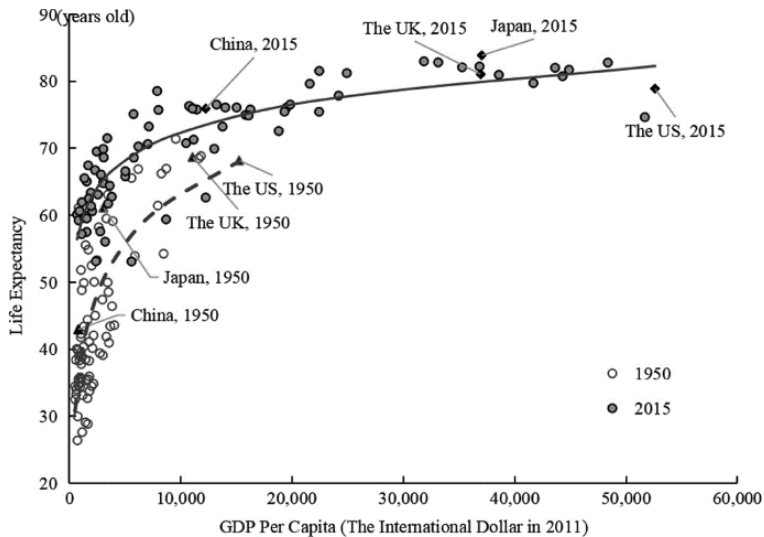
Non-linear relationships: The relationship you study is not linear.

- This is somewhat related to the non-identical observations issue. The effect may change depending on the range of a variable
- Higher income does not increase as much life expectancy for the rich as for the poor.
- More sleep is not associated with more happiness depending on if you were sleeping little or a lot already.

Identifying and solving the issue:

- This non-linear relationship is indicated by the fact your error term is not normally distributed
- The solution is usually to put some variables to the $\log()$, the square, or another simple function
- **The general idea is to fall back on a linear relationship**

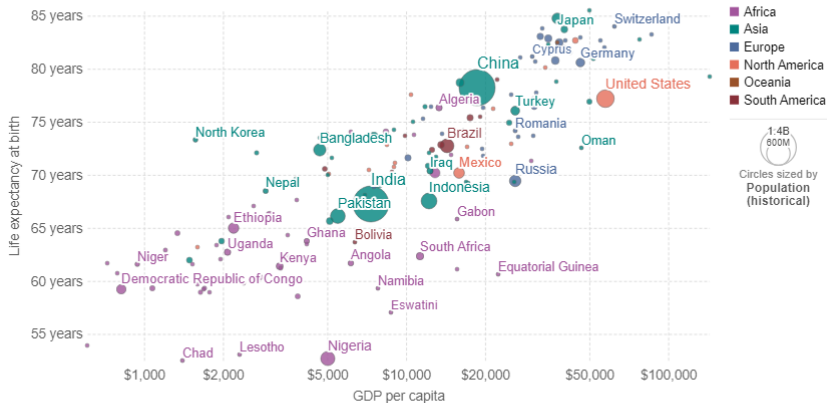




Life expectancy vs. GDP per capita, 2021

The period life expectancy at birth, in a given year. GDP per capita is adjusted for inflation and differences in the cost of living between countries.

Our World
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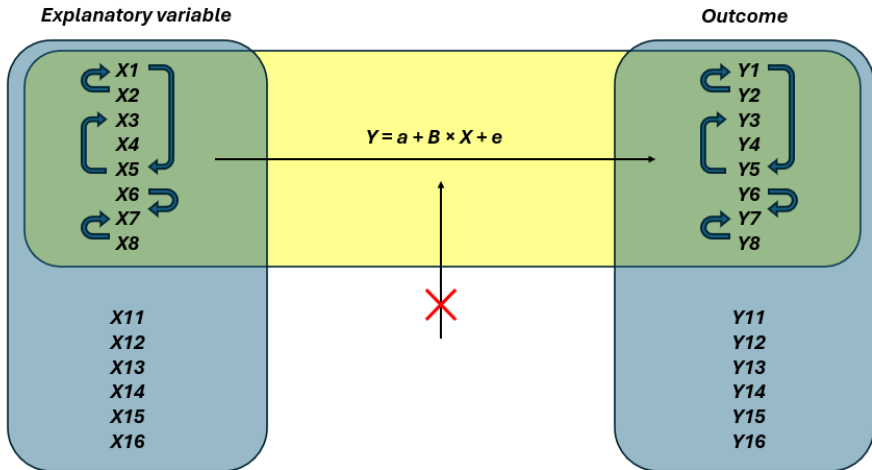
Data source: UN WPP (2022); HMD (2023); Zijdeman et al. (2015); Riley (2005); Bolt and van Zanden - Maddison Project Database 2023

Note: GDP per capita is expressed in international-\$ at 2011 prices.

OurWorldInData.org/life-expectancy | CC BY

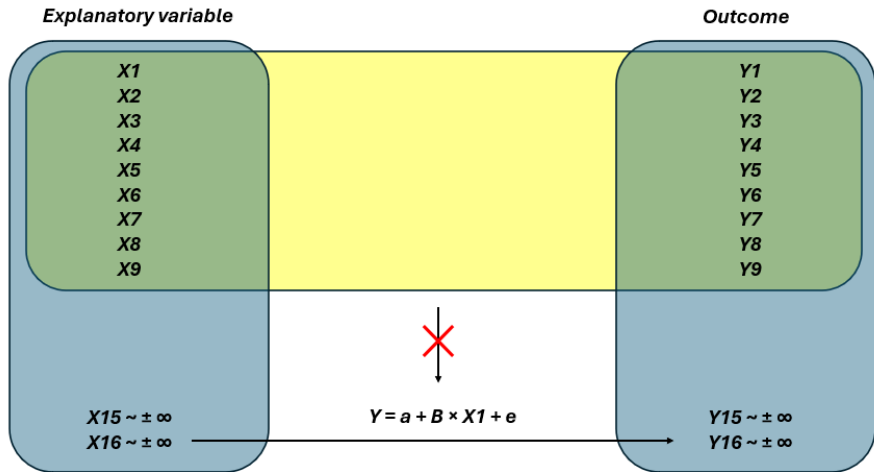
Non-independent observations: your observations interact with each other.

- You study the impact of having more holidays on student grades. You observe students. However, students from a same school will behave the same way (because they all go to the same school).
- You look at the efficiency of the vaccine in curbing a spreading viral epidemic. However you observe multiple members of a same household (the chances of contamination between them is far greater than the rest of the population)
- The solution may be to use weights (not the scope of this class)
- The solution may be to **aggregate values** (compare the school average grades and not compare students for instance)
- The solution may be to **add another control variable** (that checks if the school effect on its students grades)



Outliers

- It is somewhat related to the non-linear and non-independent issues
- Observations with very high/low values tend to drive the results
- It is common practice to remove them **if it is relevant and properly justified**
- It is common practice to remove them and see if results hold as **robustness checks**



P-hacking: This is not a data rather than a "you" problem. It is what happens when you play too much with your variables to find the perfect result

- If you have 100 variables, it may happen that, sometimes, you have significant results
- This is because a 90% or 99% confidence interval is NOT a 100% confidence interval
- [▶ Link](#)
- The solution is that your regression model should be **coherent with theory, and robust to other designs**
- **This is why robustness checks are necessary**

